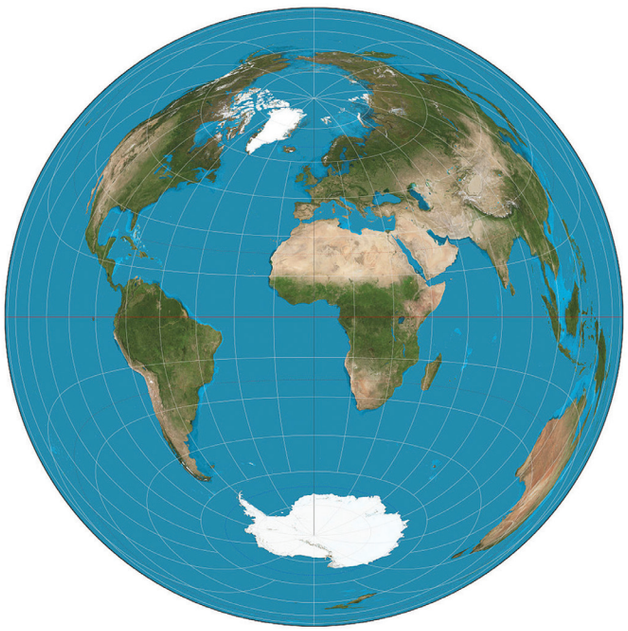
Polar Zenithal Projection

Polar Zenithal Equal area projection

This projection is invented by J.H Lambert in the year 1772. It is also known as Lambert's Equal Area Projection.



- The pole is a point forming the centre of the projection and the parallels are concentric circles.
- The meridians are straight lines radiating from pole having correct angular distance between them.
- The meridians intersect the parallels at right angles.
- The scale along the parallels increases away from the centre of the projection.
- The decrease in the scale along meridians is in the same proportion in which there is an increase in the scale along the parallels away from the centre of the projection. Thus the projection is an equal area projection.
- Shapes are distorted away from the centre of the projection. Scale along the meridians is small and along the parallels is large so the shapes are compressed along the meridians but stretched along the parallels.
- Used for preparing political and distribution maps of Polar Regions. It can also be used for preparing general purpose maps of large areas in Northern Hemisphere.

Example 10.3

Construct a Polar zenithal equal area projection for the whole globe with a reduced earth radius of 4 cm and the latitudinal and longitudinal interval 30°

Steps of construction:

- Draw a circle with radius equal to 4 cm representing a globe. Let CD and AB be the polar and equatorial diameters respectively which intersect each other at right angles at O, the centre of the circle.
- Draw radii OE, OF, OC, making angles of, 30°, 60° and 90° respectively with OB. Join CB, CE, and CF by straight lines.
- With radius equal to CF as centre draw a circle. The point represent 90^0 parallel and

TN_GOVT_XII_Geography_chapter10.indd 172 12/10/2021 11:57:49 AM